

MINGZHE LI

Household Finance · Portfolio Choice

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WORKING PAPERS

On Apparent Relative Risk Aversion (2026)

Available at [SSRN 6911380](#)

Abstract: For HARA utility satisfying the precautionary saving motive, apparent relative risk aversion—the value function’s relative risk aversion, which governs the optimal risky portfolio share—lacks a general characterization under concave consumption. Given a constant $1/(k - 1)$, it is characterized by the elasticity of supernumerary consumption with respect to wealth, bounded above by unity. Linear consumption implies that apparent relative risk aversion is increasing and concave, such that the optimal risky portfolio share is decreasing and convex in the Merton model, a pattern difficult to reconcile with limited participation and underdiversification. This paper derives necessary and sufficient conditions for the monotonicity and concavity of apparent relative risk aversion under concave consumption. These conditions provide a benchmark to assess whether deviations rationalize heterogeneous investor behavior.

WORK IN PROGRESS

Apparent Risk Aversion in the Small and in the Large

This paper concerns the value function for risky assets. A measure of apparent risk aversion in the small, the risk premium for an arbitrary risk, and a natural concept of decreasing apparent risk aversion are discussed and related to one another.

On Apparent Relative Risk Aversion: Numerical Evidence from Continuous-Time Portfolio Choice

Develops a numerical framework to evaluate the analytical conditions for apparent relative risk aversion monotonicity and concavity established in the companion paper, for problems where closed-form consumption policy is unavailable.

Household Problem with Risk Heterogeneity: A Continuous-Time Framework

To allow the cost functional to have a stochastic response to risk sources, this paper proposes a term in the HJB equation that characterizes the cross-effect between household endowment risk exposure and welfare sensitivity to risk sources. This interaction enables us to isolate the effect of heterogeneous risk on household welfare.

RESEARCH EXPERIENCE

Julia (2024) Implementation of *Financial Frictions and the Wealth Distribution* (Fernández-Villaverde et al., 2023)

MATLAB (2024) Codebase modernization for *Recursive Macroeconomic Theory* (Ljungqvist & Sargent, 2018) Available at [GitHub](#)

EDUCATION

Shihezi University 2024–Present
Master of Finance (M.F.) GPA: 85.74/100

Sichuan University 2023
B.Eng. in Computer Science and Technology

SKILLS

Programming: Julia, MATLAB

Mathematical Training:

Analysis: *Analysis I–III* (H. Amann & J. Escher)

Stochastic Analysis: *An Introduction to Stochastic Differential Equations* (L. Evans)

Languages: Chinese, English

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